Pathi Krishna Kanth

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LinkedIn | GitHub | LeetCode

Education

Pragati Engineering College — Surampalem, India
Bachelor of Technology in Computer Science and Engineering; CGPA: 8.80 — *2022 – 2026*

Experience

Full Stack Developer — Dec 2025 – May 2026
CarTrade Tech — *On-site*

- Engineered responsive web client interfaces in **Vue.js** backed by modular **PHP** micro-services to streamline platform workflows.
- Optimized database schemas and complex queries across **MongoDB** and **MySQLi**, accelerating data retrieval speeds by 35% and enforcing strict data consistency.

Projects

LightMem | *Rust, SQLite, FTS5 BM25, Vector Search, UniFFI, Embedded Systems* — GitHub

- Engineered an ultra-fast, zero-daemon agent memory engine in **Rust**, achieving sub-5ms CLI execution with embedded single-file SQLite persistence.
- Designed a hybrid retrieval pipeline uniting **SQLite FTS5 BM25** exact keyword matching with vector cosine similarity and Reciprocal Rank Fusion (RRF).

MacSystem-MCP | *Python, FastMCP, Model Context Protocol, macOS APIs, Accessibility* — GitHub

- Architected a high-throughput **Model Context Protocol (MCP)** workstation server exposing 87 native macOS system controls to autonomous LLM agents.
- Implemented OS-level automation primitives via macOS Accessibility Trees (**AXUIElement**) and CoreGraphics for programmatic UI inspection and event dispatching.

LocalShare 2.0 | *Python, TCP Sockets, AES-256-GCM, Tailscale, FastMCP, Mesh Network* — GitHub

- Engineered a high-speed encrypted mesh file-sharing protocol over raw TCP utilizing custom length-prefixed binary framing and chunked **AES-256-GCM** encryption.
- Maximized network throughput using **4x parallel socket multiplexing** with asynchronous out-of-band ACKs and real-time **Tailscale** peer mesh routing.

QueryPort Headless CMS | *React 19, TypeScript, Node.js, Express, MongoDB, REST APIs* — GitHub

- Architected a headless CMS and developer portfolio platform utilizing the **MERN stack** under a decoupled, strict MVC design pattern.
- Implemented secure RESTful endpoints with unified envelope response schemas, centralized error-handling middleware, and cloud media asset persistence.

Technical Skills

Languages: Rust, Python, TypeScript, JavaScript, Java, PHP, C, SQL
Systems & AI: Model Context Protocol (FastMCP), Vector Search, Local LLMs (Ollama), UniFFI
Web & Cloud: React.js (v19), Vue.js, FastAPI, Node.js, Express.js, Docker, Linux, REST APIs, Raw Sockets
Databases: SQLite, MongoDB, MySQL
Core Concepts: Data Structures & Algorithms, Concurrency, Computer Networks, System Design, Operating Systems

Achievements & Leadership

Competitive Programming & Academics

- **LeetCode Knight:** Solved 1,300+ algorithmic problems; consistently ranked within the top 5% globally in weekly contests.
- **NPTEL Java DSA:** Ranked in the top 1% nationally with an elite score of 87%.

Leadership & Extracurriculars

- **GDG Lead (Algorithms):** Led the algorithms track for Google Developer Group; conducted technical workshops and hosted competitive coding events.
- **1st Prize:** Won 1st place in campus-wide "Crack the Jumble Code" algorithmic problem-solving competition.